

Corporate Bond Relative Value

Curve Fitting and Duration-Neutral Portfolio Tests

Course	MTH 9897 - Systematic Trading
Instructor	Professor Misha Boroditsky
Sample period	August 2022 to March 2025
Issuers	AAPL, AMZN, BA, CAT, DIS, and T

5.99% Selected portfolio gross annualized return	3.60% Annualized return after 25 bp costs	0.05% Annualized return excluding AAPL
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Main result

The liquidity-aware portfolio produces the strongest result among the three tests, but the AAPL exclusion check shows that the improvement is not broad across issuers. The report therefore treats the backtest as a useful portfolio-construction exercise, not as evidence of a ready-to-trade strategy.

Prepared from the executed WRDS analysis notebook

Executive Summary

This assignment asks whether an issuer-level spread curve can identify rich and cheap corporate bonds, and whether those residuals can support a duration-neutral long-short portfolio. I estimate a separate weighted curve for each issuer and month, form positions at month-end, and evaluate the next available monthly return.

Key findings

- The baseline top-three strategy earns 2.75% annualized before costs. Concentrating in the top two signals raises the return to 4.48%.
- A liquidity-aware top-three rule produces the strongest result: 5.99% gross annualized return, 8.66% volatility, and a 0.69 gross Sharpe ratio.
- At a 25 bp one-way turnover cost, the selected strategy falls to 3.60% annualized with a 0.42 Sharpe ratio.
- The result is heavily concentrated in AAPL. Excluding AAPL reduces annualized return to 0.05% and Sharpe to 0.06.
- The duration scaling is mechanically effective. Net DV01 is approximately zero, but the portfolio still carries issuer, spread-curve, liquidity, and shorting risk.

16.21% Gross total return	5.99% Gross annualized return	8.66% Annualized volatility	0.69 Gross Sharpe ratio
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Interpretation

The three tests confirm that portfolio construction matters. The improvement from the liquidity adjustment is economically meaningful in this sample, especially after trading costs. At the same time, the AAPL sensitivity and the sharp final-month increase make it difficult to argue that the effect is stable. A stronger follow-up would select the rule on an earlier training period and evaluate it on later observations.

1. Data and Sample Construction

The analysis uses WRDS enhanced end-of-month corporate bond returns for six large non-financial issuers. The common sample begins in August 2022 and ends in March 2025. I require at least one year to maturity and approximately \$200 million outstanding, exclude defaulted and convertible observations, and remove implausible spread, price, and duration values.

Within each issuer-month, I retain the modal security level and rating category. This keeps the curve focused on bonds with comparable seniority and credit quality.

Issuer	Observations	Unique bonds	Median duration	Duration range	Median spread
AAPL	1,475	52	7.31	0.95 - 20.99	19.0 bps
AMZN	908	33	6.58	0.95 - 20.03	15.0 bps
BA	1,304	51	7.39	0.92 - 16.41	28.5 bps
CAT	957	51	3.45	0.95 - 18.65	13.0 bps
DIS	1,237	44	8.82	0.96 - 18.19	20.0 bps
T	1,678	56	10.63	0.96 - 17.84	28.0 bps

Return treatment

Positions are formed at month-end and evaluated using the next observation for the same issue when the gap is between 25 and 45 days. WRDS RET_EOM is the primary total-return measure. If it is missing, the fallback adds the clean-price return and accrued coupon using the actual day gap divided by 360. In this sample, the fallback is not needed for the positions that enter the backtest.

Data limitations

The supplied files do not provide a complete call schedule, bid-offer spread, 144A indicator, subsidiary mapping, or fixed-versus-step-up coupon flag. These omissions matter because they can explain persistent residuals. I leave them as limitations instead of inferring them from incomplete fields.

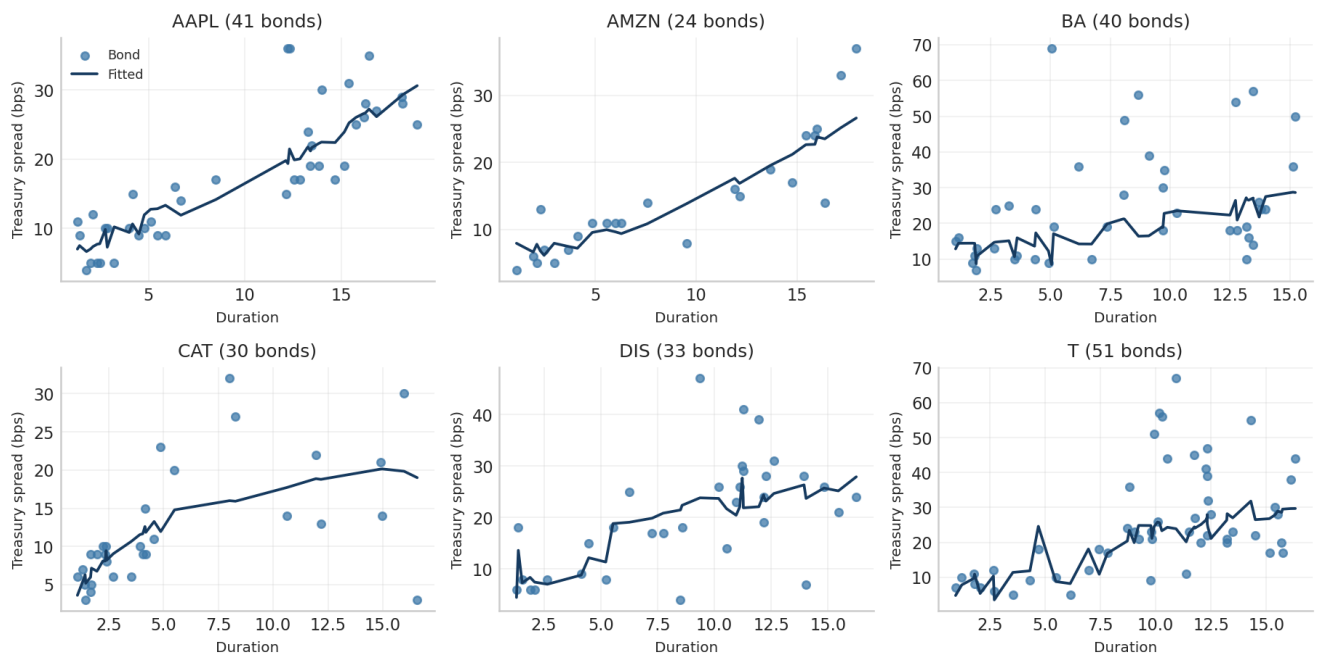
2. Issuer Spread Curves

For each issuer-month, I estimate the following weighted regression:

$$T_Spread = b_0 + b_1 \log(1 + Duration) + b_2 Duration + b_3 Coupon$$

Duration is used instead of raw maturity because it also defines the hedge constraint. Coupon helps absorb systematic cash-flow and price differences. Regression weights combine TRACE dollar volume and amount outstanding, with winsorization inside each issuer-month so one very large issue cannot dominate the fit.

Issuer spread curves at March 2025



Issuer	Fit months	Average bonds	Median weighted R2	Median RMSE
AAPL	32	46.1	0.744	6.3 bps
AMZN	32	28.4	0.755	5.3 bps
BA	32	40.8	0.376	10.5 bps
CAT	32	29.9	0.617	7.1 bps
DIS	32	38.7	0.535	8.4 bps
T	32	52.4	0.386	12.8 bps

A positive residual means the observed spread is wider than the fitted curve and is classified as cheap. A negative residual is classified as rich.

3. Portfolio Construction

All three tests buy cheap bonds and short rich bonds. The short side is scaled so the signed market-value duration contribution is approximately zero, and gross market value is then normalized to one.

	$\text{sum}(\text{Position} \times \text{Price} \times \text{Duration}) = 0$
	$\text{sum}(\text{Dollar DV01}) = 0$

Three specifications

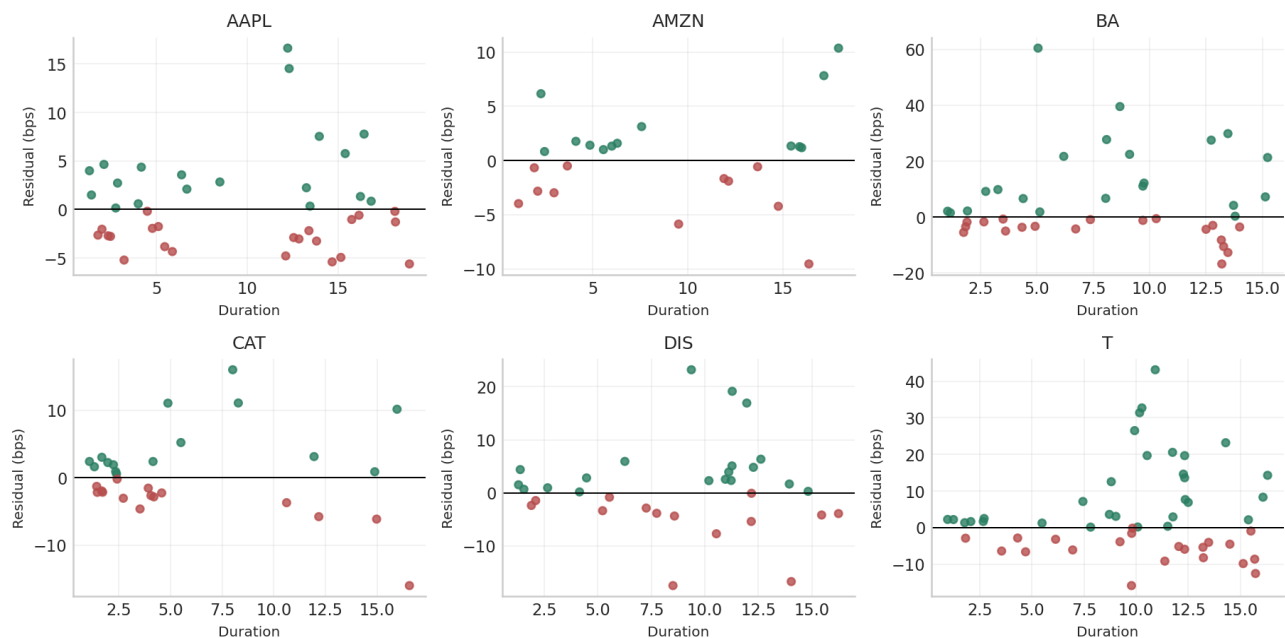
- Baseline top 3: take the three widest and three tightest residual z-scores, with position size proportional to signal strength.
- Concentrated top 2: use only the two strongest signals on each side. This may capture larger convergence moves but increases issuer-specific risk.
- Liquidity-aware top 3: multiply the residual z-score by a capped square-root measure of TRACE dollar volume relative to the issuer-month median.

The liquidity adjustment does not impose a hard liquidity cutoff. It moves more actively traded residuals up the ranking while retaining the direction of the curve signal.

Risk that remains

DV01 neutrality only addresses a first-order parallel rate move. The portfolio can still carry Treasury-curve slope and curvature risk, spread-duration risk, issuer beta, downgrade risk, and liquidity exposure. Shorting can also be costly or unavailable in practice.

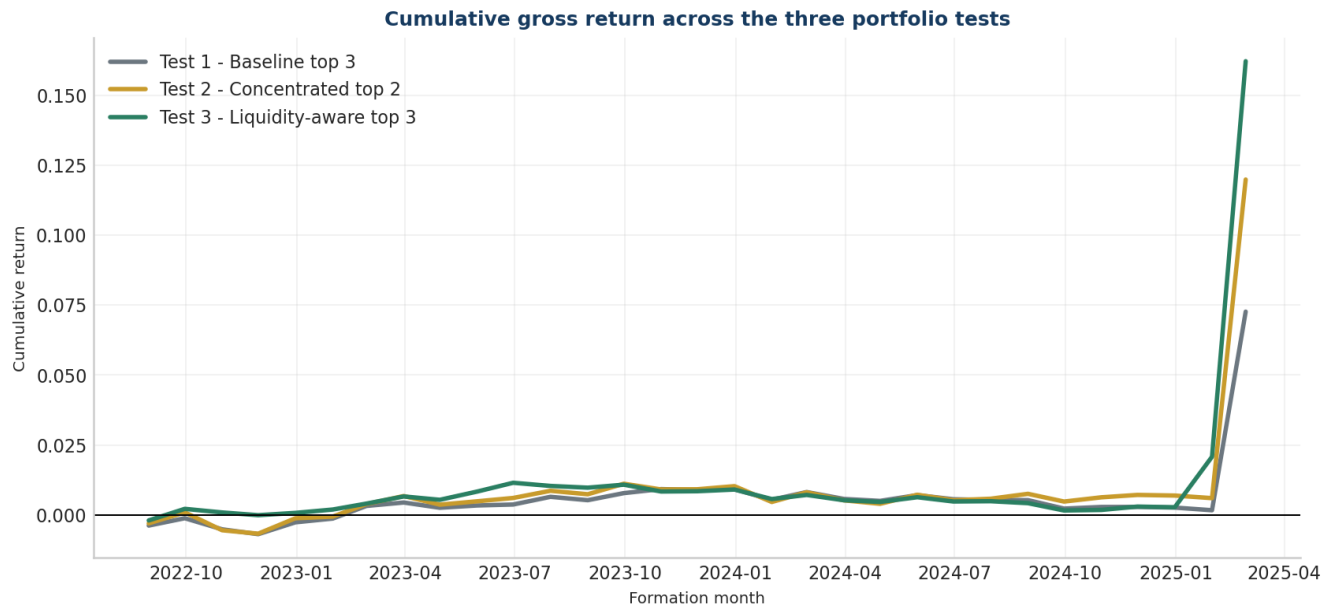
Rich and cheap residuals at March 2025



4. Comparison of the Three Portfolio Tests

The table reports the same 31-month evaluation window for all three versions. The selected portfolio is the rule with the highest Sharpe ratio after a 25 bp one-way turnover charge, rather than the rule with the highest gross return alone.

Portfolio test	Gross ann. return	Gross vol.	Gross Sharpe	Avg. turnover	Net ann. return (25 bp)	Net Sharpe (25 bp)
1. Baseline top 3	2.75%	4.47%	0.62	0.78	0.45%	0.10
2. Concentrated top 2	4.48%	7.10%	0.63	0.82	2.02%	0.29
3. Liquidity-aware top 3	5.99%	8.66%	0.69	0.79	3.60%	0.42



The liquidity-aware rule has the strongest gross and cost-adjusted result. The cumulative chart also shows that much of the separation occurs near the end of the sample, which makes the apparent improvement less convincing than the full-period return alone suggests.

Selection warning

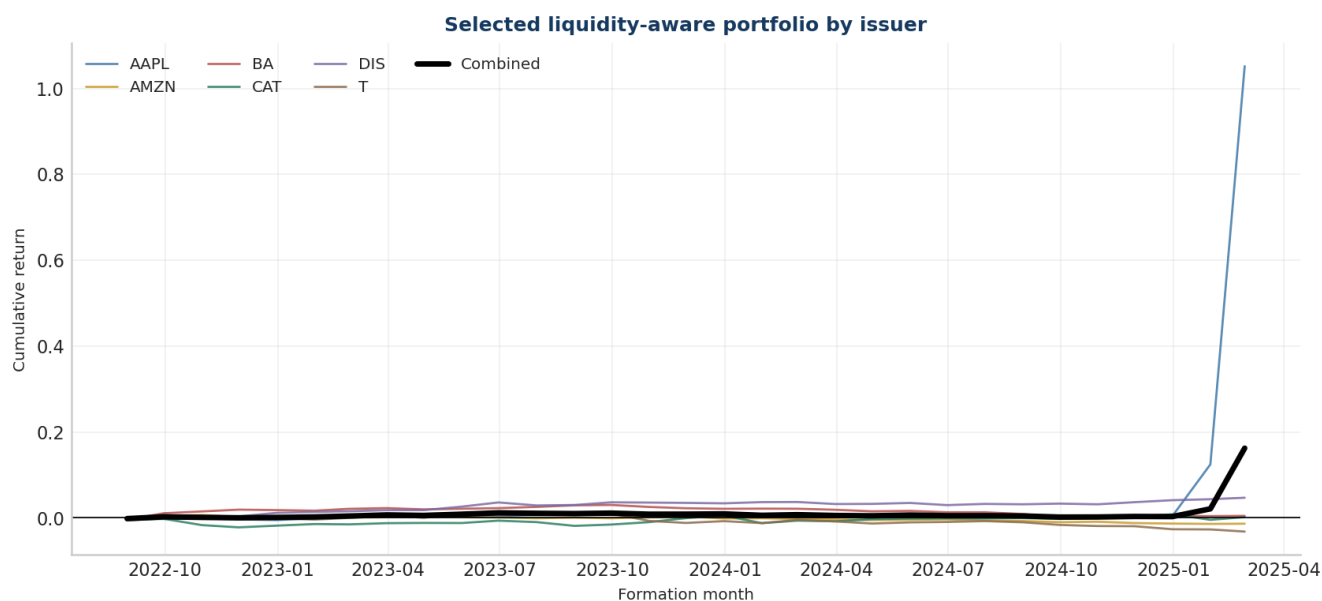
The three rules are evaluated and selected on the same sample. Choosing the winner in-sample introduces model-selection bias. A genuine out-of-sample test would choose the rule using earlier data and keep the later period untouched until evaluation.

5. Selected Portfolio Performance

The liquidity-aware top-three portfolio earns 5.99% gross annualized return with 8.66% volatility and a 0.69 Sharpe ratio. On the \$1 million gross-capital convention, the sum of monthly gross PnL is \$159,240.

5.99% Gross annualized return	8.66% Annualized volatility	0.69 Gross Sharpe ratio	\$159,240 Sum of monthly gross PnL
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Issuer	Ann. return	Ann. volatility	Sharpe	Total return	Win rate	Avg. turnover
AAPL	32.08%	51.62%	0.62	105.19%	54.8%	0.86
AMZN	-0.54%	0.67%	-0.81	-1.38%	45.2%	0.77
BA	0.17%	1.36%	0.13	0.44%	54.8%	0.75
CAT	0.06%	2.23%	0.03	0.15%	61.3%	0.77
DIS	1.78%	1.25%	1.42	4.67%	67.7%	0.84
T	-1.26%	1.78%	-0.71	-3.21%	35.5%	0.75

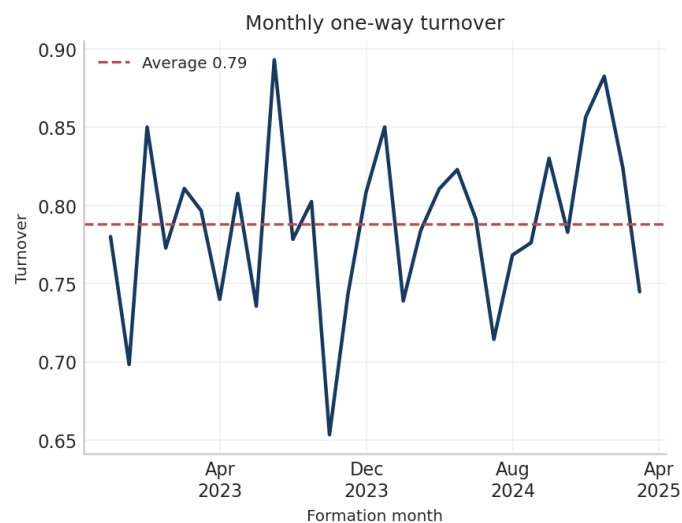
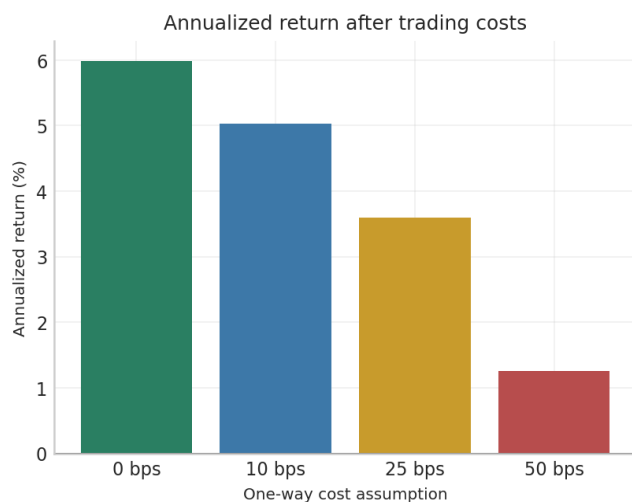


The combined result is strongly influenced by AAPL, including a 13.84% return in the best month (2025-02-28). The worst combined month is 2024-01-31 at -0.34%.

6. Trading Costs and Robustness

Corporate-bond relative-value trades can have modest expected returns relative to bid-offer spreads, borrow costs, and market impact. I therefore apply one-way cost assumptions to monthly one-way turnover.

Cost assumption	Ann. return	Ann. volatility	Sharpe	Total return	Win rate
0 bps	5.99%	8.66%	0.69	16.21%	54.8%
10 bps	5.03%	8.66%	0.58	13.51%	41.9%
25 bps	3.60%	8.66%	0.42	9.56%	22.6%
50 bps	1.26%	8.67%	0.15	3.28%	9.7%



Issuer concentration

Sample	Ann. return	Ann. volatility	Sharpe	Total return
All six issuers	5.99%	8.66%	0.69	16.21%
Excluding AAPL	0.05%	0.77%	0.06	0.12%

Removing AAPL reduces annualized return from 5.99% to 0.05% and Sharpe from 0.69 to 0.06. This is the strongest caution in the backtest. The portfolio rule improves the headline return, but the gain is not broadly shared across the remaining issuers.

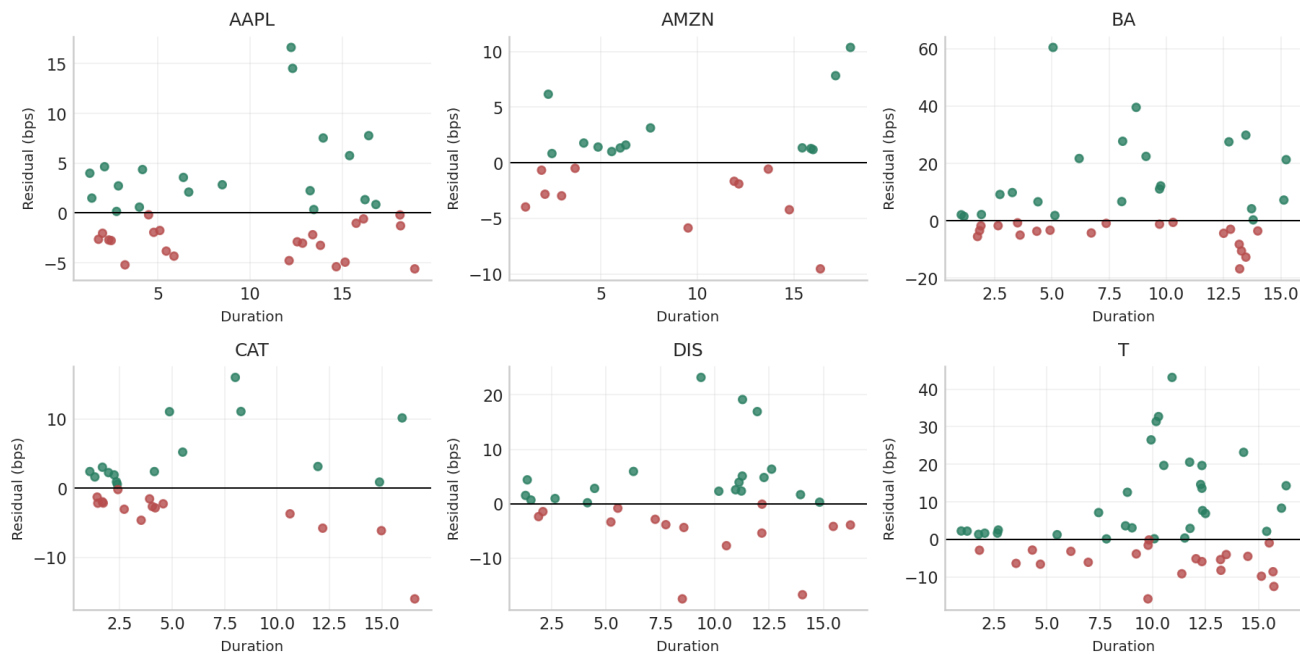
7. Outlier Review

Professor Boroditsky discussed several reasons a bond can remain away from an issuer curve: liquidity, issue size, age, coupon structure, optionality, capital-structure position, and differences between the named issuer and the legal issuing entity. The supplied fields allow a partial check of these explanations.

Selection group	Observations	Median abs. residual	Median dollar volume	Median amount outstanding	Median duration	Median TMT
Long cheap	558	22.3 bps	\$35,557,574	\$750,000	12.32	18.95
Not selected	6,443	3.9 bps	\$45,349,053	\$1,000,000	6.86	8.24
Short rich	558	10.4 bps	\$109,797,947	\$1,750,000	13.21	23.75

The long-cheap observations have materially lower median dollar volume and smaller amount outstanding than the non-selected observations. They also have longer duration and time to maturity. This pattern suggests that part of the cheap signal compensates investors for liquidity and long-dated risk rather than representing a pure pricing error.

Rich and cheap residuals at March 2025



8. Conclusion

The assignment provides a useful test of issuer-curve relative value. Weighted curve fitting produces economically interpretable residuals, and the duration scaling removes first-order parallel-rate exposure as intended. The three portfolio tests also show that ranking and concentration decisions can materially change the backtest.

The liquidity-aware rule has the best result in this sample, including after a 25 bp turnover charge. I would not interpret the 5.99% gross annualized return as a stable expectation, however. Most of the improvement comes from AAPL and from the end of the sample. The long-cheap group is also less liquid and longer-dated than the bonds that are not selected.

My conclusion is therefore narrower: the residual signal is useful for organizing relative-value opportunities, but the apparent return is sensitive to portfolio design, issuer concentration, and trading frictions. The next step would be a training/test split with point-in-time security attributes, bid-offer estimates, and borrow constraints.

Assumptions and limitations

- The three portfolio rules are selected and evaluated on the same sample.
- The hedge neutralizes parallel-rate DV01, not all Treasury-curve or spread risk.
- Trading-cost cases do not include financing, borrow fees, failed fills, or asymmetric market impact.
- Call schedules, 144A status, subsidiary distinctions, insurance, and step-up coupons are not fully observed.
- Corporate bonds can be expensive or impossible to short, so the portfolio is a research construction.

References

Boroditsky, M. MTH9897 Quantitative Investment Framework, Lectures 1-2, 2026.

Nelson, C. R., and Siegel, A. F. (1987). Parsimonious Modeling of Yield Curves. *Journal of Business*, 60(4), 473-489.

WRDS Corporate Bond Returns, Enhanced End-of-Month files supplied for the course assignment.

Wharton Research Data Services. WRDS Bond Returns documentation:
<https://wrds-www.wharton.upenn.edu/pages/get-data/wrds-bond-returns/wrds-bond-returns/>